Artificial Intelligence: An introduction

Fabio Martínez, Ph.D - Gustavo Garzón, MSc.

Escuela de ingeniería de sistemas e informática
Biomedical **I**maging, **V**ision and **L**earning **L**aboratory. **BivL^{2}ab**
Machine learning **A**nalysis and **C**omputer **V**ision. **M$_L$ACV**
Universidad Industrial de Santander

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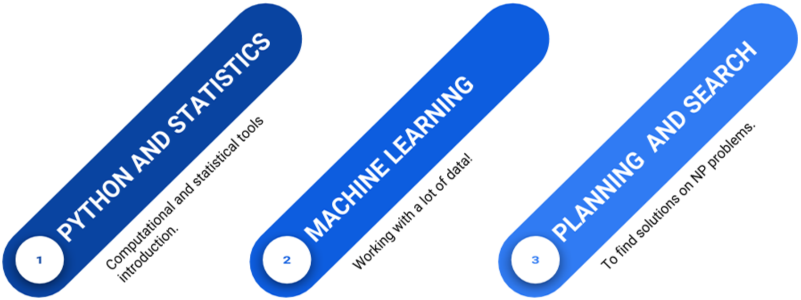
Información del profesor

- Nombre: Fabio Martínez Carrillo
- Correo (contacto y comunicación): famarcar@saber.uis.edu.co

- Nombre: Gustavo Garzón Villamizar
- Correo (contacto y comunicación): gustavo.garzon@saber.uis.edu.co

- Contenido del curso: `https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

Course outline



- **The course** is mainly based on the **coding of I.A concepts** and the **development of small applications**
- **The course** is focused on the analysis and study of **I.A algorithms**

I.A. Course

Course

It is about learning the fundamental tools to face problems and challenges of I.A. as well as some computational tools and theoretical concepts.

First module: Python and statistics introduction

- Introduction to Python
- Introduction to statistics
- Introduction to Visualization and data analytics

Second Module: Machine Learning

Clasification and regression

- Naive Bayes
- Classification trees
- Support Vector Machines
- Introduction to Deep-learning

Non-supervised learning

- K-means
- DBScan

Third module: Planning and Searching

- Planning and searching
- Genetic algorithms (GA)
- Simulated annealing (SA)

Structure of the Course

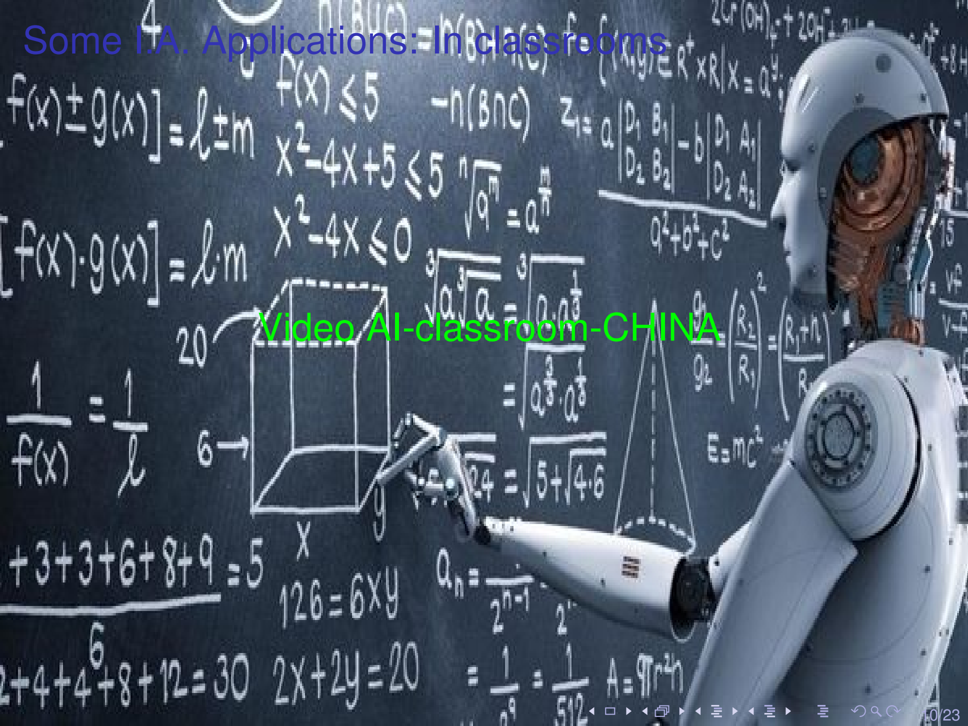
- 30% Talleres (Problemsets)
- 10% Ejercicios clases. Las notas serán de acuerdo a los cuartiles de participación del curso. Q3=5, Q2=4, Q1=3. Para los que hayan estado por debajo de Q1 se les promediara las notas de los ejercicios con respecto al total.
- 30% Parciales (Quizes)
- 30% Proyecto funcional IA.
  - 5 % primera entrega. Exploración de datos
  - 5 % segunda entrega. Validación de modelos clase
  - 20 % Entrega final. Incluye repositorio, presentación y avance del proyecto

Curso en GITLAB:
`https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student`

PREVIOUS PROJECTS

- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2018-2.md
- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-1.md
- https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2019-2.md
https://gitlab.com/bivl2ab/academico/cursos-uis/ai/ai-uis-student/-/blob/master/projects/Proyectos_2020-1.md

Some I.A. Applications: In classrooms

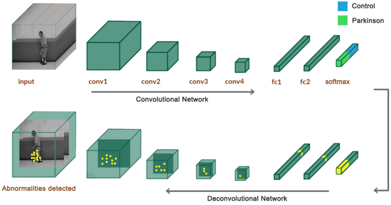


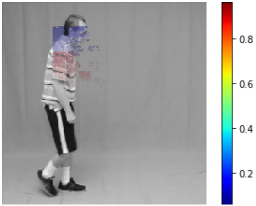
Video AI-classroom-CHINA

Some I.A. Applications: In Agriculture

Video AI-AGRICULTURE

Salient Gait maps

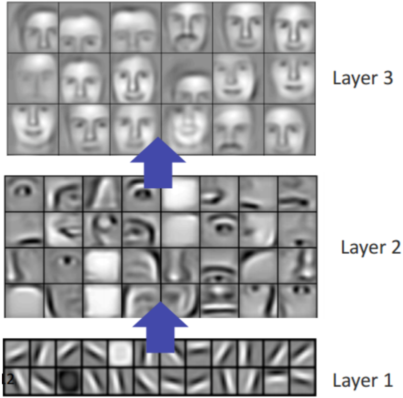


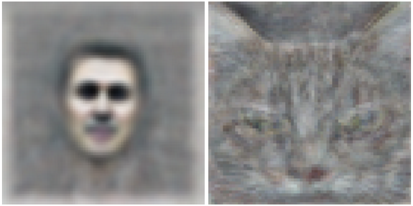


Learning convolutional neuronal patterns

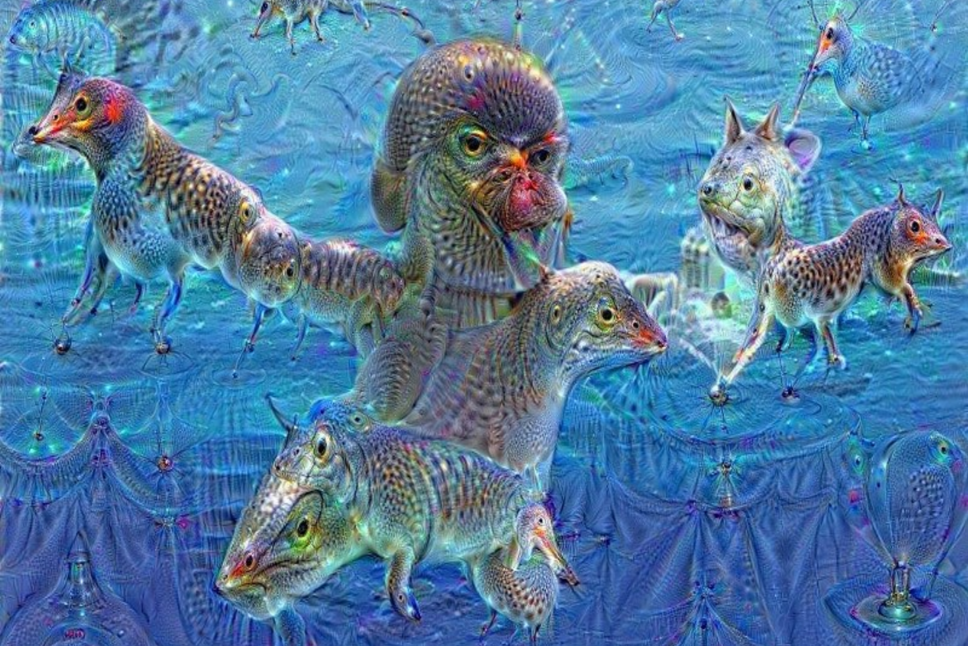
A 3D convolutional gait representation that produces salient spatio-temporal regions can potentially support and complement the diagnosis and the following of Parkinson's disease

Big challenges: How understand complex concepts?

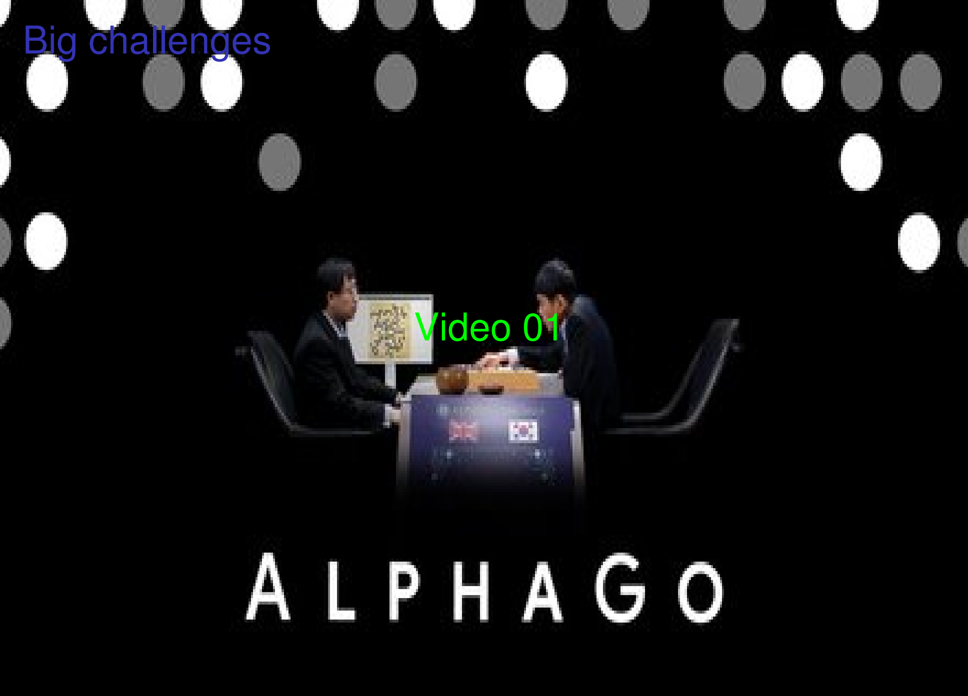




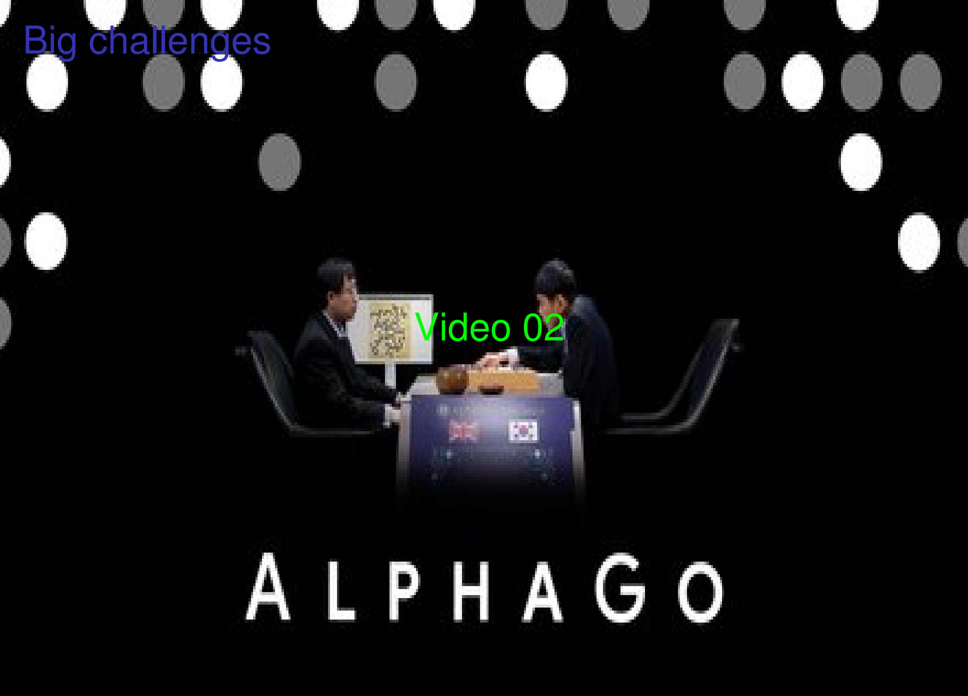
Now, the machines can dream



Big challenges



Big challenges



ImageNet Challenge

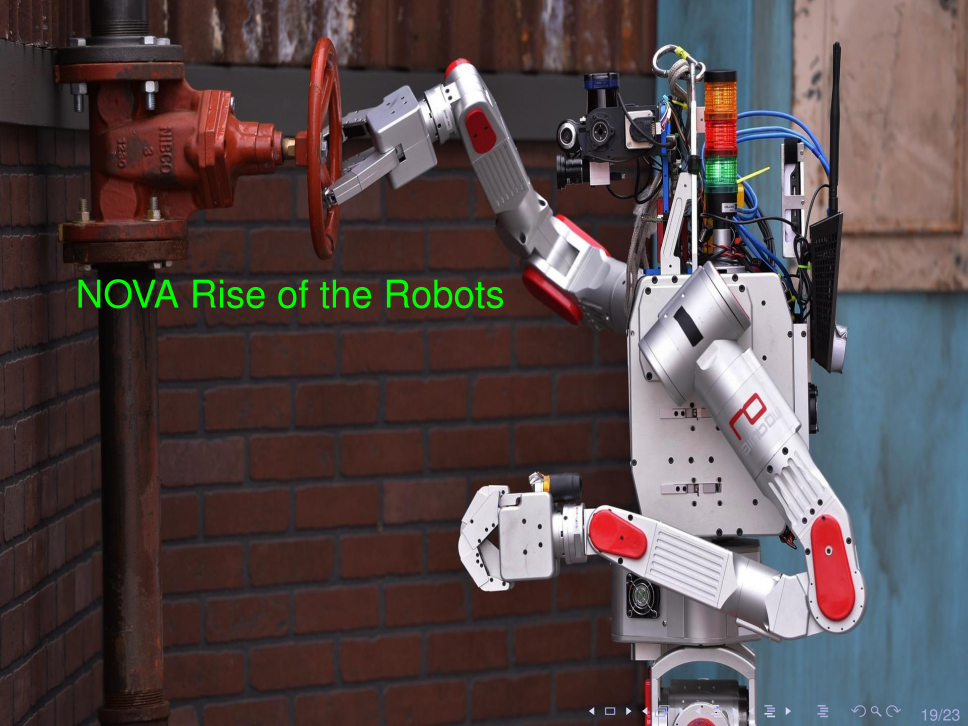
IMAGENET Netflix 22:30. Bill Nye

- 1,000 object classes (categories).
- Images:
  - 1.2 M train
  - 100k test.

As support for decisions making

- Medicine
- Politics
- Laws

NOVA Rise of the Robots



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Understanding Artificial Intelligence and Its Future

Thank you for your attention . . .

. . . It’s time to wake up

Universidad Industrial de Santander

MACV Motion Analysis and Computer Vision

BIVL²ab Biomedical Imaging, Vision and Learning Laboratory